



Numerical study of Heat Modes of laser welding of dissimilar metals with an intermediate insert



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ABSTRACT

A 3D model of heat transfer in inhomogeneous materials taking into account phase transitions for analysis of laser welding of dissimilar metals using intermediate insert is developed. Based on the proposed model, an algorithm is constructed and numerical studies are carried out of the distribution of temperature fields in the weld joint of titanium and stainless steel with an intermediate copper insert. The processes of melting, evaporation, and solidification of the materials depending on the beam power, its speed, and position of the focal spot in the system are studied. The welding modes reducing interaction of elements of the materials welded and, thus, preventing formation of brittle intermetallic phases in the weld joint are found.

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1. Introduction

Welding of titanium-based alloys and iron, including bimetallic ones (titanium alloy – stainless steel), is an important technological process in the manufacturing of parts and mechanisms that have wide application in aerospace, aviation engineering, power engineering, and shipbuilding. The problem of joining titanium and steel, particularly of austenitic class, by melting using a laser beam is yet to be solved in practice, despite the numerous studies on joining dissimilar metals by rolling, friction, explosion, etc. [1–11].

The direct welding of titanium and steel does not give positive results [6]. According to the state diagram of titanium–iron, the solubility of iron in α -titanium is extremely small and is within 0.05–0.1% at the normal temperature. At concentrations of Fe higher than 0.1%, brittle intermetallic compounds of types TiFe, TiFe₂, Ti₂Fe, and eutectics of various compositions are formed in the alloy, which drastically reduce the plastic properties of the material. Therefore, one of the main problems in obtaining a durable welded steel and titanium joint is the selection of the welding materials, methods and modes of the welding, in which the probability of creation of the brittle phases would be prevented or significantly decreased. To obtain good quality welds when welding

titanium alloy with stainless steel, a promising solution is the use of intermediate layers of other alloys. For example, in [6], for this purpose, various metal inserts (copper, bronze, etc.) were used. The most effective was the insertion of a copper plate, which allowed us to obtain joints with the tensile strength of about 350 MPa.

The laser welding takes place in a small area exposed to laser radiation, causing high temperature with large gradients and forming a weld pool. If the laser power is above a critical value, the liquid metal in the vicinity of the beam axis boils to form vapor-gas channel (keyhole or pore) of small dimensions, from which the metal vapors escape at high speed. Because of the presence of metal inhomogeneity, laser instability, hydrodynamic instability of the keyhole surface, the neighborhood of the contact between the beam and metal is dynamically unstable and surrounded by dense, bright metal vapor cloud (vapor plume). These and other circumstances in the laser welding make it very difficult to measure the physical parameters of the process and the visual observation of the welding zone. However, the processes taking place in this small-sized area determine properties and quality of the weld joint obtained. The difficulties associated with the measurements of some parameters and certain instability of the complex laser welding process complicate the creation of a flexible technology and its application. Therefore, in view of the demand for laser welding in various areas of the industry and some of its advantages over other types of welding, its numerical simulation

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Nomenclature

A, A_e	absorption factor, equivalent absorption factor	x, y, z	Cartesian coordinates
c_i^j	specific heat capacity of i th phase of j th metal	$x(z)$	keyhole generatrix
e_z	ort of z -axis	$Z(x, y)$	surface of the keyhole in x, y, z coordinates
g	acceleration of gravity	Z_c	coordinate z of a point on the keyhole surface
H	depth of the keyhole	Z_F	coordinate z of the beam focus
h_m	maximal linear size of the mesh cell	α_{k1}	coefficient of convective heat transfer as a result of gas blowing
h_z	coordinate of symmetry point of the keyhole wall generatrix	α_{ri}^j	coefficient of radiation heat transfer for metal j in phase i
K_c	curvature of the keyhole surface	ε_i^j	reduced degree of metal j blackness in phase i
l	characteristic length of the cooling zone	θ	tangent of inclination angle of the keyhole wall generatrix $x(z)$ to z -axis
L_V	latent heat of the metal vaporization	κ^j	melting heat of j th metal
$\dot{m}$	mass rate of the metal evaporation	λ_0	wavelength of the laser radiation
$\mathbf{n}$	(n_x, n_y, n_z) weld pool surface unit normal vector	λ_g	the gas heat conductivity
q	heat flux;	λ_i^j	heat conductivity of i th phase of j th metal
q_b, q_r, q_v, q_c, q_b	heat fluxes on the keyhole wall: absorbed, re-reflected, lost on metal evaporation, convective, the model correction, respectively	ν_g	kinematic viscosity of the gas
P	pressure	ρ_3	the average density of melt in the pool
Pr, Re	Prandtl and Reynolds numbers	ρ_i^j	density of i th phase of j th metal
R	gas constant	σ	coefficient of surface tension of the liquid metal
R_c	radius of spherical part of the keyhole bottom surface	σ_0	Stefan–Boltzmann constant
r_F	radius of the beam in the focal plane	Ω, Ω_{ct}	computation domain and domain of the keyhole
S_b	area of side surface of the keyhole		
S_c	sum of areas of the keyhole outlet and side surface		
T	temperature		
T_{av}	the average temperature on the keyhole wall in the laser spot		
T_m	melting temperature		
T_{sat}	boiling temperature of the alloy		
t	metal plate thickness		
U_g	flow rate of the inert gas		
(u, v, w)	vector of a particle velocity in the melt with components u, v, w		
V_k	is the volume of molten metal number k in the weld pool		
V_{tot}	is the total volume of all molten metals in the weld pool		
$\mathbf{V}_w$	velocity of the beam progress (welding rate)		
W	laser beam power		

Subscripts

c	vapor-gas channel (keyhole)
g	gas
i	index of i th phase
r	recoil reaction
s	static
sat	boiling
m	liquid phase
v	vapor

Superscripts

av	averaged
j	index of j th metal

is of current interest. According to the mathematical model that adequately describes the properties of the process under study, it is possible to calculate the parameters of interest from the technological perspective. In this context, the numerical simulation can significantly complement the physical experiment, which is often much more expensive than computer calculations [12–18].

In this paper we consider the process of butt welding of plates made of two different metals with a thin intermediate insert between them of a third metal (see Fig. 1). Our goal is to construct an adequate mathematical model for the calculation of temperature fields, location of the phase boundaries of melting and solidification in the welded metals depending on the thickness and material of the insert, position of the focal spot of the laser beam, the radiation power and welding speed. This model can be further used for determining the optimal welding conditions excluding (or reducing) the formation of undesirable intermetallic compounds.

In this paper, the 3D quasistationary thermophysical model proposed earlier and described in detail in [12–13] is generalized to the case of welding of two dissimilar plates with an insert of a third metal.

Difference of this model from the one in [12–13] is that for each metal, melt in the weld pool, and solidified welding joint it has different thermal parameters in all the equations and boundary

conditions. For example, the coefficients of thermal conductivity and heat capacity, heat of fusion, and other parameters. Due to the convective mixing, the weld pool contains the mixture of molten metals. Therefore, averaged values of the parameters are used in this subdomain. The averaging procedure is described in Section 2.2 in detail.

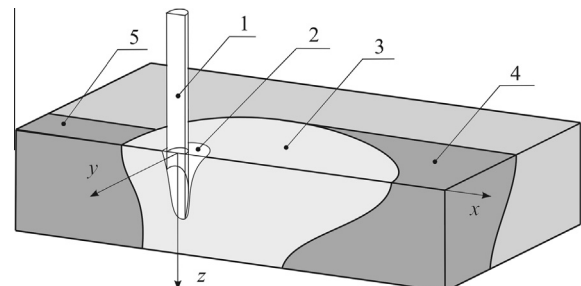


Fig. 1. Welding scheme. 1 – laser beam, 2 – vapor-gas channel (keyhole), 3 – melt pool, 4 – solid phase, 5 – intermediate insert.

2. Mathematical model overview

In laser welding, several physical processes take place simultaneously, creating a complex picture, the model of which in interaction of its components is still far from being complete. From the viewpoint of thermophysical modeling, it can be assumed in an approximation that almost all the energy emitted by the laser is located within a beam with radius of about 0.1 mm in the cross-section and that the radiation density has a Gaussian distribution along the radius. Experimenters note that diameter of the input cross-section of the keyhole is slightly larger than the diameter of the beam cross-section, but less than twice [14].

For numerical simulation of the complex process of metal plates laser welding, researchers have used several models taking into account different number of constituting it simpler physical processes that have different degrees of influence on the results. In order to simplify their description in the formulation of the model and make obtaining quantitative results possible, one has to make a number of simplifying assumptions.

Disturbances in the movement of the liquid metal on the keyhole walls cause high-frequency low-amplitude oscillations in the liquid metal. They propagate from the keyhole throughout the pool, significantly decaying along the way due to the melt viscosity. Here, we consider the steady process with constant welding speed, laser power, and other parameters. In this case, pulsations of parameters of the gas and liquid metal flows in each small volume in the moving coordinate system associated with the laser beam occur near their average values. Therefore, the average values of the process parameters at each point of the computational domain in the welding zone and at the periphery can be considered constant with a high accuracy.

In the simplest thermophysical model, in order to calculate the temperature field, the heat transfer from laser to metal is described by a moving linear heat source in a semi-infinite medium. In more sophisticated models, the following factors are taken into account in various combinations: presence of the keyhole in the weld pool in the vicinity of the laser beam axis, the power density distribution along the radius of the beam, the presence of the liquid and solid phases in the welding area. The models can also take into account the absorption of laser radiation heat and its partial reflection from the surface of the metal, the heat discharge due to the metal evaporation from the keyhole surface, the radiation absorption and heat carried away by metal vapor flowing out of the pore, the thermal radiation from the plates, contact heat transfer from them to the mounting table and gas contacting it in the welding area, and other factors. All factors mentioned here will be also taken into account to some extent in the model adopted here. Some details will be described below in the appropriate sections.

We consider a steady-state continuous laser butt welding of two plates of dissimilar metals with the insert different in composition from the materials welded (see Fig. 1). The plates have the shape of rectangular parallelepipeds with equal thickness. During the welding, the laser beam axis lies in the plane parallel to the junction of the plates, is perpendicular to their upper surface, and moves in the direction opposite to x -axis. The plates are blown by the inert gas to protect the liquid metal from oxidation.

In the domain under consideration, we introduce Cartesian coordinate system x, y, z , in which the laser beam is stationary, while the plates move with the welding speed $\mathbf{V}_w = (V_w, 0, 0)$. We assume that the welding speed is constant, while the temperature field and positions of the phase boundaries are quasi-stationary. Axis z is oriented downward along the beam axis, axis x goes along the junction of the plate with the insert in the direction of the plates movement, and axis y is perpendicular to the joint. The coordinate origin is located on the beam axis on the upper surface of

the plates welded. Fig. 1 shows the welding domain located in half-space $y < 0$. It contains one of the plates welded. Through the insert it is tightly connected with the other plate welded, which is located in half-space $y > 0$ and is not shown in the figure for the clearness of the whole picture. With this method, the molten pool 3 contains a mixture of all three metals, which are the materials of the plates welded and of the insert.

The front wall of the melt pool (with respect to the direction of the beam movement) adjoins a thin region with the metals in two-phase state. It was replaced with a surface in the present study under Stephan model approximation. This is reflected in Fig. 1. Stephan conditions are set on front and rear walls of the pool in the solidification and melting zones in the calculation of heat transfer. The pool through the rear wall is adjacent to weld joint 5 formed by the solidified alloy.

The weld pool is elongated in the direction of beam movement along the metal surface. Its volume depending on the amount of heat absorbed by the metal, the cross-sectional area, as the experience of modeling and comparison of calculations with the physical experiment show, can be easily calculated assuming that all the metal particles in the pool move in a plane-parallel way. The rapid transfer of heat in it due to natural convection of the melt can be modeled by an increased coefficient of the thermal conductivity in the region occupied by the pool. Here, we do not take into account the influence of the convective movement of the melt in the pool on its shape, in contrast to how it was done in [13,18].

Analyzing the results obtained there, it should be noted that the convection of the melt in the pool affects to some extent the pool form in the longitudinal direction and slightly in the transverse and vertical directions. Therefore, for calculating the thickness and width of the weld and determining which surfaces of the plates melted, this method of modeling of the heat transfer in the pool gives acceptable results. Based on the thermophysical model without using viscous fluid dynamics equations, one can satisfactorily calculate certain parameters of the welding process: the size of the weld pool, the weld joint width, the time of the metal particles stay in a certain temperature range, as well as to predict the grain size in the crystalline structure of the solidified melt [8–10,12].

2.1. Simulation of the keyhole

Various models of heat transfer from the laser beam to the metal during the welding are proposed [12–14,17,18]. Their different advantages and disadvantages that are not discussed here. In most of them, at first, the hole in the liquid metal, created under influence of the excessive pressure (recoil pressure balanced by forces of surface tension and hydrostatic pressure) is calculated under assumptions made in a particular model. Then, taking into account the heat flux from the beam to the keyhole walls, the field temperature in the plate is calculated, ignoring the reverse effect of the temperature field in the plate on the keyhole shape. In the keyhole model used in this paper, this interaction is taken into account by the simultaneous, iterative calculation of the keyhole and temperature field in the whole computational domain.

As in [12], to simplify the model, the keyhole wall is sought in the form of a surface of revolution from a chosen parametric family. Among these parameters are the keyhole depth, angle of its wall inclination to z -axis at the point of the most intense laser radiation, and other variables. Choice of the parameters (choice of the surface of this family) is based on the experimental data and the requirement that the temperature on most part of the surface in the vicinity of the beam axis is close to the temperature of the boiling point of the metal, slightly exceeding it (5–10%) in the laser spot. The keyhole is created automatically. During the iterative process, the parameters of its surface are selected simultaneously

with the calculation of temperature fields in the entire computational domain. The algorithm for construction of the keyhole was also described in [13].

There are a few important circumstances related to laser welding of metal plates, which are taken into account in a numerical simulation depending on which characteristics of the process are to be calculated. Note that the radius of the laser beam is by more than a decimal order of magnitude smaller than the size of the weld pool, the weld joint width, and the keyhole depth. The metal vapors are discharged in the direction of the beam axis from the keyhole outlet at a rate exceeding $\mathbf{V}_w$ by more than three orders of magnitude, where $\mathbf{V}_w$ is speed of the beam movement along the plates joint (welding rate). The recoil forces from the escaping vapors are largely directed along the axis of the beam. At the actual welding parameters, this fact determines the superiority of the keyhole size in this direction over its transverse dimension.

On the keyhole walls, the forces from the momentum of the metal particles evaporating from their surface, surface tension, excess vapor pressure and recoil from of their discharge from the keyhole, and the hydrostatic pressure within the molten metal are mutually balanced. The latter one is estimated to be negligible compared with other forces at a small thickness of the plate and a small radius of curvature of the keyhole surface.

Surface tension increases with the growth of the surface curvature. Therefore, it assists the instantaneous disappearance of the folds, edges, and other features on the keyhole surface, which can occur at its small oscillations. We assume that the shape of the keyhole walls averaged over time is smooth. This is the kind of shape that is sought in this model of the quasi-stationary welding mode.

In view of the smallness of the cross section of the keyhole, the temperature, the surface tension on the front and rear parts of the keyhole surface, and the recoil force differ insignificantly. At the given ratio of $\mathbf{V}_w$, the velocity of the metal vapor particles, and keyhole transverse size, the excess pressure of the vapors in each cross-section of the keyhole is mostly leveled. At a fixed keyhole height, these forces act almost uniformly on the perimeter of the keyhole cross-section. Therefore, in a quasi-stationary mode at small Peclet numbers considered here ($Pe = vr_F/a \sim 10^{-2} \cdot 10^{-4}/(5 \cdot 10^{-6}) \sim 0.2$) [14], cross-section of the keyhole is close to a circle. Surface tension and excess pressure largely prevent the low vapor recoil forces from distorting the cross-sectional shape “averaged over time”. After many researchers [14,17,18, etc.], the keyhole cross-section is considered to be a circle.

Note also some experimental facts taken into account in the keyhole model adopted here. The experiments on laser cutting of not thick metal plates show that the cross section of holes obtained is close to the round shape and has a radius just slightly exceeding radius r_F of the beam in the focal plane. For example, in [14], its value is taken equal to $1.7 \cdot r_F$. Here, we take it equal to $2 \cdot r_F$. The difference between the last two values is usually small compared with other sizes of the keyhole, weld pool, and the product size. Numerical experiments performed by the authors have shown that it has minor effect on the shape and dimensions of the welding joint and location of key isotherms.

When moving the beam at a constant speed, the beam axis is slightly ahead of the keyhole axis and is on the front (relative to the beam movement direction) keyhole wall [11]. We assume here that the laser beam axis is ahead of the keyhole axis by r_F [17].

In the present study we do not take into account the frictional force caused by vapors flowing out of the keyhole and acting on the keyhole walls. In addition, we make simplifying assumptions listed below.

- (1) The parameters of melt in the pool also fluctuate around their mean values that are taken for the values of the parameters of the quasi-stationary process calculated.

- (2) The surface of the keyhole is a surface of revolution about an axis parallel to axis z and lying in the plane of symmetry, monotonically narrowing with the depth 10,14,18, etc.
- (3) The bottom of the keyhole locally is a spherical surface with radius R_c .
- (4) The volume of liquid metal displaced from the keyhole is neglected here because it is less than 1% of the weld pool volume. The deformation of the weld pool surface caused by this displacement and other factors are not taken into account in the present model. In other words, the top surface of the weld pool is assumed to be flat coinciding with the top surface of the plates. It introduces an error comparable with the accuracy of available input data for thermophysical constants.

Neglecting the hydrostatic pressure of the liquid metal, we determine the value of R_c from the conditions of local forces equilibrium caused by the vapor pressure and surface tension according to formula

$$R_c = \sigma/p, \quad (1)$$

Here, σ is the melt surface tension, p is the excess vapor pressure which is the sum of the static pressure at the surface evaporation p_v and recoil pressure p_r :

$$p = p_v + p_r, \quad (2)$$

Following [16], we determine p_r by relation

$$p_r = 0.55p_v. \quad (3)$$

The highest temperature on the surface of the keyhole is located near the axis of the beam and cannot significantly exceed the boiling point of the melt. Among all the possible generatrices of the keyhole surface (which, according to the second assumption, is a surface of revolution), the curve lying in the plane $y = 0$ on its front wall will be the most “warmed up” (i.e., according to the accepted model, of all the generatrices, it will have the longest part, where the boiling point is reached). Therefore, we take it as the generatrix of the keyhole surface. At the construction of this curve, we will be guided by the following assumption: the temperature should be close to the boiling temperature of alloy T_{sat} on its maximum possible part near the laser beam axis for the given parameters of the welding.

Generatrix AC of the keyhole surface, lying in plane $y = 0$, is sought here in the form of spline: a curve consisting of two parts AB and BC ($AC = AB \cup BC$). AB is a cubic polynomial tangentially mating BC , which is a part of circle with radius R_c (see Fig. 2b). At that, it is advisable to consider AB belonging to two-parameter family

$$M(\theta, h) = \left\{ x(z) : x(z) = \left(\frac{r_F - \theta h}{h^3} \right) (z - h)^3 + \theta(z - h), 0 \leq \theta \leq r_F/h, h > 0 \right\} \quad (4)$$

with independent parameters θ and h . The curves of this family under the assumptions adopted pass through point A (Fig. 2b), and their parts are symmetric with respect to the point of their intersection with line $\{x = 0, y = 0\}$, which is the axis of the laser beam. At that, the parameter h is z -coordinate of point of symmetry AB , and θ is the slope of curve $x(z)$ with z -axis at the point of symmetry (the minimum angle of inclination of curve AC to z -axis). Curve $x(z)$ is symmetric with respect to the point of symmetry only before point B . That is, $x(z)$ is symmetric only in the vicinity of axis of the laser beam. At point B , curve $x(z)$ fits a portion of a circle, which radius R_c is estimated above. The surface of the keyhole with generatrix AC constructed in this manner satisfies the assumptions 2–4. Note, by the way, that the location of the beam relative to the front wall of the keyhole is in a good agreement with the one

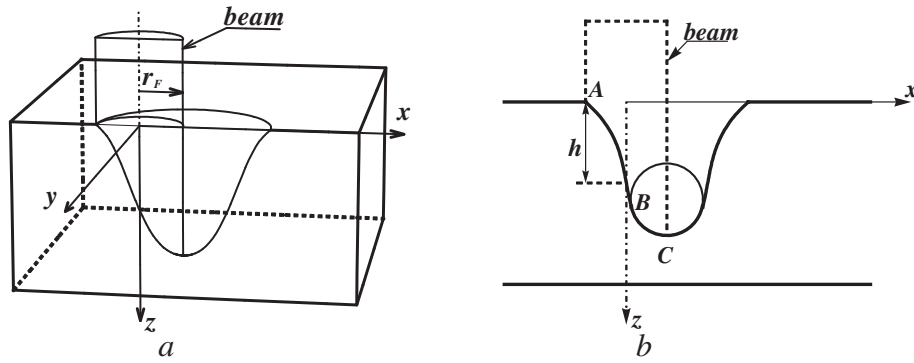


Fig. 2. Position of the laser beam relative to the keyhole (a) and the shape of its generatrix (b).

chosen in [11], and the shape of the wall generatrix is in a good agreement with the shape of front wall of the keyhole as shown there too.

Suppose that we can solve the problem of finding the temperature field in the plate at a known and fixed shape of the keyhole surface. In this case, the problem of finding this surface can be reduced to the following problem: varying the shape of the keyhole surface, it is necessary to choose the solution (i.e., the temperature distribution), when the temperature is close to boiling temperature T_{sat} of the alloy on the maximum part of AC possible near the axis of the laser beam. In other words, controlling parameters θ and h , successively, iteratively, simultaneously with the temperature calculation in the product, we try to construct the best curve AC from the perspective of the principle formulated. This is done in the manner described below.

2.1.1. Governing by parameter h

We calculate the average temperature T_{av} on section AB symmetric with respect to the point of intersection with z -axis and located near the axis of the laser beam. If $T_{av} \gg T_{sat}$, then the melt is “overheated” and it is necessary to increase the depth of the keyhole calculated (this is achieved by increasing the parameter h). The angle between the beam axis and the normal to surface of the keyhole increases, and so does the area of the keyhole surface absorbing the direct laser radiation. These factors lead to a decrease of temperature on the surface of the keyhole. Similarly, if $T_{av} < T_{sat}$, h needs to be reduced. This will raise the temperature on the keyhole surface. This is how the depth of the keyhole is adjusted.

2.1.2. Governing by parameter θ

After choosing an appropriate h , it may happen that at the intersection of AC and the laser beam axis, the temperature exceeds (or, alternatively, does not reach) boiling point T_{sat} , while at a short distance from AC on both sides of the beam axis, the opposite picture is observed. This testifies to a wrong choice of parameter θ characterizing the angle of the keyhole wall inclination to z -axis. Reducing (or increasing) the angle of AC curve inclination, we make sure that the nearest neighborhood of crossing point of AC curve and the laser beam axis gets less (more) heat compared to the periphery. (This is due to the fact that the power density of the laser radiation has a Gaussian distribution, and the heat absorbed by the keyhole walls from the direct laser radiation is proportional to sinus of angle of inclination of the keyhole wall to the laser beam axis). Thus, by varying θ and h , we find the optimal surface shape of the keyhole.

The further refinement of the shape of the keyhole wall [13] can be carried out taking into account the conditions of the heat balance and dynamic equilibrium

$$p(z) = \sigma K_c + g \rho_2^{av} z, \quad z = Z_c. \quad (5)$$

Here, K_c is the curvature of the keyhole, Z_c is z -coordinate of a point on the surface, ρ_2^{av} is the average density of the alloy in the weld pool. Due to the small size of keyhole cross sections in the directions perpendicular to z axis, and the strong impact of surface tension forces at large K_c , up to the terms of higher order, the keyhole surface can still be considered as a surface of revolution (for the same reasons, the shape of the lower part of the keyhole bottom is close to a part of a sphere). Therefore, the calculation of keyhole shape can be reduced to construction of its generatrix.

Pressure $p_v(z)$ can be expressed as a function of temperature using a simplifying hypothesis. Finally, two remaining unknown quantities $T(z)$ and $x(z)$ can be found by the iterative calculation of the temperature distribution in the plates and the adjustment of the keyhole surface.

2.2. Main equations and relations

Quasi-stationary process of heat transfer in the weld pool and solid metal in a moving coordinate system is described by equation

$$c_i^j \rho_i^j v \frac{\partial T}{\partial x} = \lambda_i^j \frac{\partial^2 T}{\partial x^2} + \lambda_i^j \frac{\partial^2 T}{\partial y^2} + \lambda_i^j \frac{\partial^2 T}{\partial z^2}, \quad (6)$$

where c_i^j , λ_i^j , ρ_i^j are specific heat capacity, coefficient of thermal conductivity, and density of i th phase ($i = 1$ for the solid phase, $i = 2$ for the liquid one) of j th metal. This paper describes the process of welding plates of titanium ($j = 1$) and steel ($j = 2$) with a copper insert ($j = 3$).

The weld pool actually contains the mixture of molten metals ($i = 2$). The weld joint adjacent to the rear wall of the weld pool contains an alloy of metals of the solidified melt ($i = 1$).

To represent this fact in the model, the thermophysical parameters in the liquid phase for all metals were set to averaged values according to the contribution of each metal:

$$c_i^{av} \rho_i^{av} = \frac{1}{V_{tot}} \sum_{k=1}^3 c_i^k \rho_i^k V_k, \quad \lambda_i^{av} = \frac{1}{V_{tot}} \sum_{k=1}^3 \lambda_i^k V_k, \\ \rho_i^{av} = \frac{1}{V_{tot}} \sum_{k=1}^3 \rho_i^k V_k, \quad V_{tot} = \sum_{k=1}^3 V_k, \quad (7)$$

where V_k is the volume of metal number k in the weld pool and in the weld joint. At that, for the alloy ($i = 1$) on the right side (6), we take for simplicity average values of the thermophysical parameters of metals in the temperature range of the welding joint cooling. Calculation of the alloy composition in the pool is discussed below in Section 4.

There is a thin two-phase zone between the weld pool and solid zone. In order to simplify the model it is represented as a surface where Stefan condition is specified, namely:

$$\left(\lambda_2^j \frac{\partial T}{\partial n}\right)_2 - \left(\lambda_1^j \frac{\partial T}{\partial n}\right)_1 = \rho_1^j \kappa^j \mathbf{n} \cdot \mathbf{V}_w, \quad (8)$$

where $\mathbf{n}$ is the unit normal to the phase boundary, κ^j is heat of melting (crystallization) of j th metal, $-(\lambda_1^j \partial T / \partial n)_1$ and $-(\lambda_2^j \partial T / \partial n)_2$ are heat fluxes calculated from the solid and liquid phases, respectively, $j = 1, 2, 3$ in the melting zone. On the boundary between the weld pool and weld joint subscript j is replaced with av . It corresponds to averaged parameters. This approach is commonly used in the temperature-based models [4].

The boundary between the solid and liquid phases in j th metal is defined here by the melting (solidification) isotherm ($T = T_m^j$). On the border between different metals j_1 and j_2 in phases i_1 , i_2 , respectively, the condition of the heat fluxes balance is required:

$$\lambda_{i_1}^{j_1} \frac{\partial T}{\partial n} = \lambda_{i_2}^{j_2} \frac{\partial T}{\partial n}. \quad (9)$$

A boundary value problem for heat Eq. (6) is considered here in the region between planes $z = 0$ and $z = t$ with the keyhole with curvilinear boundary removed:

$$\Omega = \{(x, y, z) : -\infty < x/t < \infty, -\infty < y/t < \infty, 0 < z/t < 1\} \setminus \Omega_{ct}, \quad (10)$$

where t is thickness of the plates, Ω_{ct} is the inner region of the keyhole (see Fig. 1). In the zone of the laser radiation (on the surface of the keyhole $z = Z_c(x, y)$, where $(x - r_F)^2 + y^2 \leq (2r_F)^2$), the heat balance equation is valid

$$-\lambda_2^{av} \frac{\partial T}{\partial n} = \mathbf{q}_l \cdot \mathbf{n} + q_r + q_v + q_c + q_b, \quad (11)$$

where $\mathbf{n} = (n_x, n_y, n_z)$ is unit normal to the surface of the keyhole, directed into the region of the liquid phase, $\mathbf{q}_l = q_l \mathbf{e}_z$ is the absorbed laser radiation flux without rereflection, $\mathbf{e}_z$ is ort of z -axis, q_r is heat flux emerging from multiple reflections from the keyhole walls, q_v is heat loss due to the metal evaporation, q_c is a term that simulates the convective heat flux from the front keyhole wall to the rear one, q_b is a correction introduced to compensate for the fictitious flux of heat transferred through the surface of the keyhole due to simplifying assumptions of the model.

The metal vapors escaping the keyhole create a vapor plume which affects the amount of energy from the laser beam entering the welded plates. It can absorb quite amount of energy from the laser beam. To simplify the model, the effect of vapor plume was simulated by adjustment of the coefficient of absorption. Taking into account the complexity of experimental measurement of coefficient A , we used the values published in [15], where it was varied from 8% to 17% depending on the welding rate and beam power. In the calculations presented in this paper the best fit with experimental data was observed at $A = 8\%$.

The welding is performed by radiation of CO₂ laser with a wavelength $\lambda_0 = 10.6 \mu\text{m}$, the radiation intensity is described by a normal Gaussian distribution. Density of the laser radiation flow is given by [18]:

$$q_l(x, y, Z(x, y)) = \frac{2WA}{\pi r_F^2} \exp\left(-2 \frac{x^2 + y^2}{r_F^2}\right), \quad r_z = \left[r_F^2 + \left(\frac{z - Z_F}{\pi r_F \lambda_0}\right)^2\right]^{1/2}, \quad (12)$$

where r_F is radius of the beam in the focal plane, W is the laser power, Z_F is z -coordinate of the beam focus, A is coefficient of absorption of the laser radiation by the keyhole surface. For simplicity, we assume that the rest of the beam power after the first and further reflections from the keyhole walls $W_i = (1 - A) W_{i-1}$ ($i = 1, 2, \dots$) is uniformly dispersed over the surface S_c consisting of

the keyhole side surface S_b and its outlet at $z = 0$. Then, we calculate the multiply reflected radiation power absorbed by the keyhole walls, using the absorption equivalent coefficient

$$A_e = A \sum_{n=1}^{\infty} \left(\frac{(1 - A)S_b}{S_c}\right)^n = \frac{A(1 - A)S_b}{S_c - (1 - A)S_b} \quad (13)$$

by formula $W_r = A_e W$. In the last equation, S_b , S_c are areas of the corresponding surfaces.

For the power density of the reflected laser radiation absorbed by an element of surface of the keyhole, we obtain expression

$$q_r = \frac{W_r}{S_b} = \sum_{n=1}^{\infty} \left(\frac{(1 - A)S_b}{S_c}\right)^n = \frac{A(1 - A)W}{S_c - (1 - A)S_b}. \quad (14)$$

Heat balance Eq. (11) takes into account the heat loss due to evaporation of the metal. To estimate it, the model uses formula

$$q_v = -L_v \dot{m}, \quad (15)$$

where L_v is latent heat of the metal vaporization, $\dot{m}$ is the mass evaporation rate per unit surface of the keyhole defined by equation

$$\dot{m} = \sqrt{p_v \rho_v}, \quad (16)$$

where ρ_v is the density of metal vapors in the keyhole. Pressure p_v was calculated using Clausius–Clapeyron relation written in integral form

$$p_v/p_{vm} = \exp[K_{tr}(T_{sat} - T_m)/RT_m]. \quad (17)$$

Here, T_{sat} and p_{vm} are boiling temperature and saturation pressure of vapors at melting temperature T_m respectively, K_{tr} is Trunton's constant (88 J/(mole·K)), R is the universal gas constant. It was taken into account in the calculations that the focal point of the laser beam is located at the boundary between the welded plated with the copper insertion. Since copper has lower melting temperature than those for steel or titanium, the formation of the keyhole occurs in the copper insertion. Therefore, the vapors in the keyhole primarily consist of copper atoms. They can be approximately considered as single-component ideal gas.

As was mentioned above, components of the melt flow velocity in the weld pool in this version of the model are assumed to be $u = V_w \rho_1 / \rho_2$, $v = 0$ m/s, $w = 0$ m/s, i.e., all particles of the melt move in plane-parallel. This simplifying assumption in the model leads to the fact that through any surface element dS of the keyhole, there is a non-zero heat flux equal to $n_x^{av} \rho_2^{av} T V_w dS$. In reality, however, these fluxes do not exist, as the liquid metal does not flow through the keyhole surface (the metal mass loss at evaporation is neglected due to its smallness), but rather flows around it, transferring heat from the well heated by the laser front wall located in the half-space $x < r_F$ into the region adjacent to the rear wall. In order to compensate for the contribution of these fictitious heat fluxes, “emerging” because of the simplifying assumptions of the model, the following term is added to Eq. (11):

$$q_b(x, y) = -n_x c_2^{av} \rho_2^{av} T(x, y, Z_c(x, y)) V_w. \quad (18)$$

To account for the convective heat transfer from the front to the rear wall of the keyhole, we introduce the following term into the right side of (11):

$$q_c(x, y) = -c_2^{av} \rho_2^{av} (T(x, y, Z_c(x, y)) - T(2r_F - x, y, Z_c(x, y))) V_w, \quad (19)$$

where $(x, y, Z_c(x, y))$ and $(2r_F - x, y, Z_c(x, y))$ are points of the keyhole surface symmetric relative to plane $x = r_F$.

The boundary condition on surface $z = 0$ in the areas of solid and liquid phases has the form

$$\lambda_i^j \frac{\partial T}{\partial z} \Big|_{z=0} = (\alpha_{k1} + \alpha_{r_i}^j)(T - T_g) \Big|_{z=0} - q_l, \quad (x - r_F)^2 + y^2 > 4r_F^2, \quad (20)$$

where T_g is the gas temperature, α_{k1} is the convective heat transfer coefficient determined in the case of gas laminar flow according to [19] by formula $\alpha_{k1} = 0.664 \text{ Pr}^{1/3} \text{ Re}^{1/2} \lambda_g / l$, where $\text{Re} = U_g l / \nu_g$, $\text{Pr} = \nu_g / a_g$, U_g is velocity of the inert gas flow, l is the characteristic length of the cooling zone, ν_g , a_g , λ_g are kinematic viscosity, temperature conductivity, and thermal conductivity of the gas, respectively. The second factor in (20) α_{r1}^j is associated with heat loss of the hot body radiation to the environment and is defined by

$$\alpha_{r1}^j = \varepsilon_i^j \sigma_0 (T^2 + T_g^2)(T + T_g), \quad (21)$$

where ε_i^j is reduced degree of blackness of metal j in phase i , σ_0 is Stefan–Boltzmann constant.

On the lower surface of welded plates ($z = t$), there is a condition similar to (20):

$$-\lambda_i^j \frac{\partial T}{\partial z} \Big|_{z=h} = (\alpha_{k2} + \alpha_{r1}^j)(T - T_g) \Big|_{z=h}. \quad (22)$$

At a sufficiently large distance from the laser beam, the temperature throughout the thickness of the plates is taken equal to ambient temperature (T_g):

$$T|_{x \rightarrow \pm\infty} = T_g, \quad T|_{y \rightarrow \pm\infty} = T_g. \quad (23)$$

3. Numerical algorithm

For numerical solution of the problem, a well-regarded conservative version of the method of collocation and least residuals (CLR) [20,21] is used. With it, the temperature distribution in the plates is determined, position of the internal phase boundaries on the basis of the temperature values obtained in the computational domain is found. Iteratively, in the process of approximate solution convergence, at the same time with the temperature field,

the keyhole surface is calculated, geometrical parameters of which (the keyhole depth, the wall inclination) are selected according to the requirements formulated above.

The calculations were carried out on irregular grids refined in the vicinity of the keyhole. Fig. 3 shows the image of the computational grid and the way of its refinement. In the grid used, the number of cells per unit length near the keyhole is substantially larger than in the areas remote from the welding zone. For clarity, Fig. 3 shows only the grid cells that are in region $\{-1 \text{ mm} \leq x \leq 2 \text{ mm}, -1 \text{ mm} \leq y \leq 1 \text{ mm}, 0 \text{ mm} \leq z \leq 4 \text{ mm}\}$. A significant part of the computational domain and the cells at the periphery are not shown in the figure. The version of the CLR method used here is accurate up to $O(h_m^2)$ when the thermophysical parameters are continuous and the computational domain is a parallelepiped. The accuracy of the method is $O(h_m)$ in the case when the thermophysical parameters have discontinuities in the computational domain and/or some boundaries of the domain are curvilinear. Here, h_m is maximal linear size of the mesh cells. The method of grid refinement used here allows of avoiding essentially elongated cells, which increase the error of discretization and numerical solution of the problem. However, it allows of using significantly fewer cells on the periphery than in the case of solving the problem on a uniform grid. This makes it possible, with limited resources (the computer memory, the time for solving the problem), to cover a significant peripheral region through which the welding zone interacts with the environment.

Instead of conditions at infinity described in Section 2, Dirichlet conditions are used here, set on a sufficiently large distance from the keyhole (20 cm). The non-uniform grid allowed us to implement the condition at infinity with high accuracy by calculating the temperature in the considerably large area. To verify this approach and control its accuracy when solving this problem, additional numerical experiments were carried out. At that, the

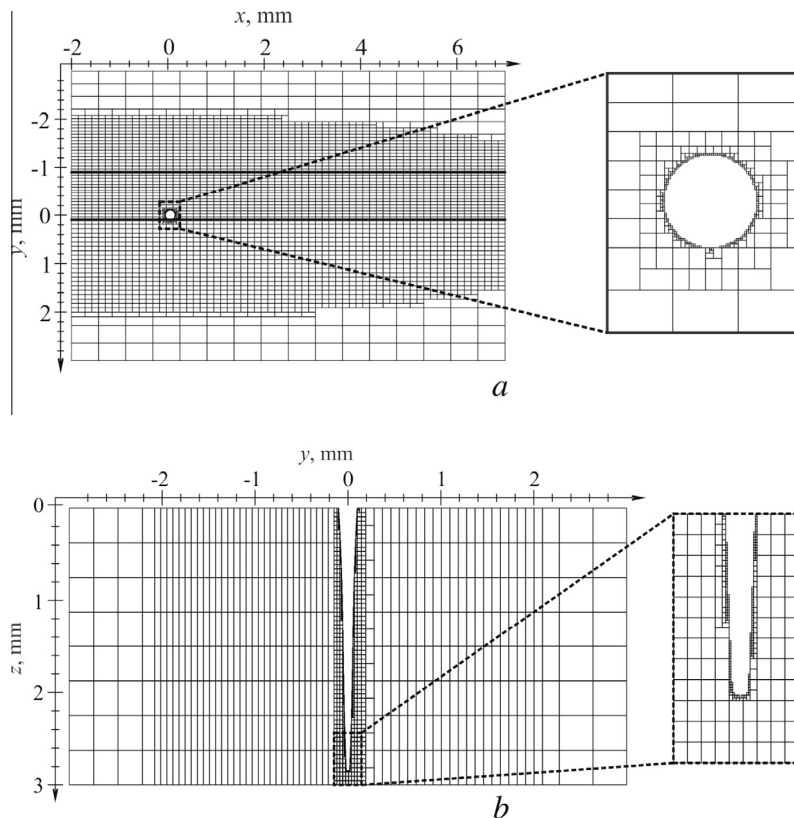


Fig. 3. Adaptive computational mesh in the weld zone, refined in the vicinity of the keyhole.

physical parameters (isotherms location, shapes of the keyhole and weld pool) of numerical solution, obtained in the first experiment in the area with a selected size in the cross-section, were compared with those of the solution built in the area with twice as less size in the cross-section. It was found out that the difference of the two solutions was insignificant.

The numerical algorithm used here is also partially described in [12–13].

In this work, we checked the accuracy with which the approximate solution satisfies the integral equation of heat balance written for the entire region Ω . To find the temperature distribution in Ω , the total heat fluxes through the surface of the keyhole and the other boundaries of Ω (heat loss through the top, bottom, and side surfaces of the plates) were calculated. The equality of heat inflow from the beam into Ω and total losses was checked. Such a control was needed to debug the program and confirm that the numerical method used allows obtaining an approximate solution satisfying the integral conservation law. Also, this test allowed us to formulate simplifying assumptions and boundary conditions in the model so that the heat balance in the problem solution is preserved. In the calculations, the results of which are presented in this paper, the approximate solutions constructed satisfy the equation of heat balance written for the computation domain Ω with a relative error not exceeding 1% of the amount of heat released to Ω through walls of the keyhole.

4. Simulation results

Below, are presented the results of numerical modeling of butt welding of plates of titanium alloy VT1 – 0 and steel 12Kh18N10T with thickness $t = 3$ mm with a copper insert with thickness of 1 mm between them. The laser power $W = 2.4$ kW, the welding speed $V_w = 1$ m/min. The temperature fields, position of internal boundaries in the weld joint, shape and depth of the keyhole are found in the numerical experiments.

Effective values of thermophysical parameters of the mixture of liquid metals in the weld pool were calculated assuming that the alloy approximately consists of 55% of copper and 45% of steel (or titanium) according to the location of the laser beam. E.g., if the laser is directed to the joint between copper insertion and titanium plate, then the volume fraction of the titanium in the weld pool is about 45%. The volume content of the third metal does

not exceed 3%. Its impact on the thermophysical parameters is insignificant. These numbers were estimated from the calculated shape and position of the weld pool.

Fig. 4 shows positions of the isotherms corresponding to the melting points of: 1 – titanium, 1,941 K; 2 – steel, 1,775 K, 3 – copper, 1,356 K, and index 4 marks isotherm with temperature of 1,100 K. In Figs. 4(b, c) and 5b, the straight lines in the middle of the computation domain denote boundaries of the copper insert before the welding. This part of the computation domain is labeled with ‘Cu’. Parts of the computation domain occupied by titanium and steel before the welding are labeled with ‘Ti’ and ‘St’. The calculation results shown in Fig. 4 correspond to the case when the laser beam focal point lies on the upper surface of the plates ($Z_F = 0$ mm). The titanium plate is in region $\{y < -0.9$ mm}, the copper insert is in strip $\{-0.9$ mm $\leq y \leq 0.1$ mm}, the steel is in region $\{y > 0.1$ mm}. The laser beam impacts the copper plate at the interface boundary with the steel. Here and below, for clarity, only a part of the computation domain is shown, containing the isotherms under consideration. The black region containing point (0,0,0) in Fig. 4(a,b), corresponds to the keyhole. The values of the thermophysical parameters used in the calculations are taken from known sources [22–24] and others.

For comparison, Fig. 5 shows the cross-sectional morphology of the welds obtained in a physical experiment (a) and in the numerical calculation (b). Note that judged from the position of the isotherms, it is not difficult to determine the cross-sectional area of melted part of each metal and calculate their share in mixture in the weld pool and weld.

One can see from Figs. 4c and 5b that at the copper–steel border, both plates are thoroughly melted during the welding, while at the titanium–copper border, only the copper insert melted over its entire height, and only a thin layer of the titanium plate melted in its upper part. Joining of its lower part with the copper insert basically was achieved through the soldering. This can be seen from comparison of the weld joint morphology calculated and experimental. Noticeable morphological difference between the calculated shape of the weld joint (right) boundary and the experimental one, obviously, is caused by neglecting in the model of the melt flows in the weld pool, affecting the local heat transfer conditions in the pool [13,17]. The volume value and qualitative picture of the penetration area of the joint are in a satisfactory agreement with the experiment.

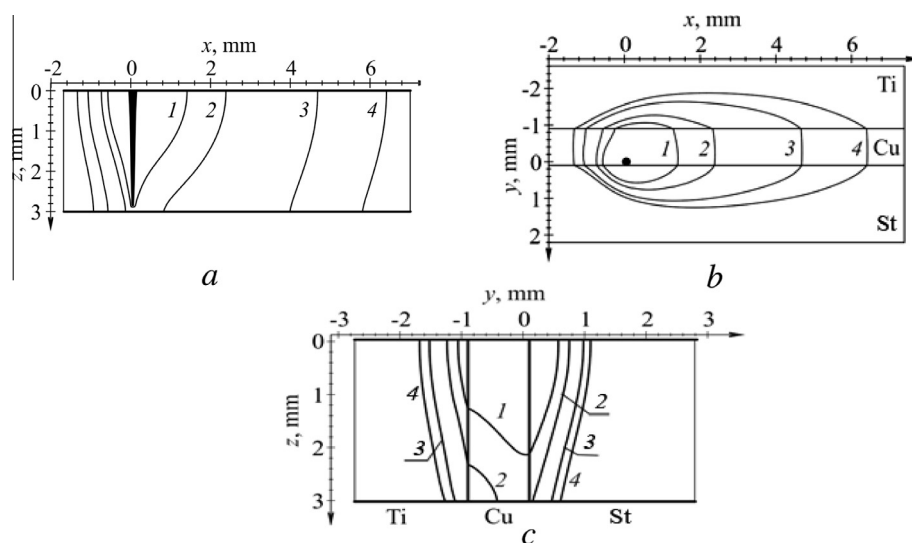


Fig. 4. Position of isotherms in cross-section $y = 0$ of the computation domain (a), on the upper surface of the plates ($z = 0$) (b), and in cross-section $x = 0.4$ mm of the computation domain (c).

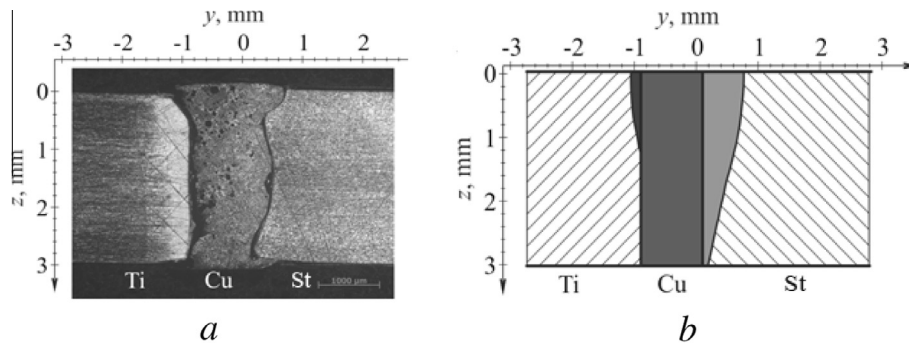


Fig. 5. Comparison of cross-sections of the weld obtained in an experiment (a) and in the calculation (b) (shown in different shades of gray).

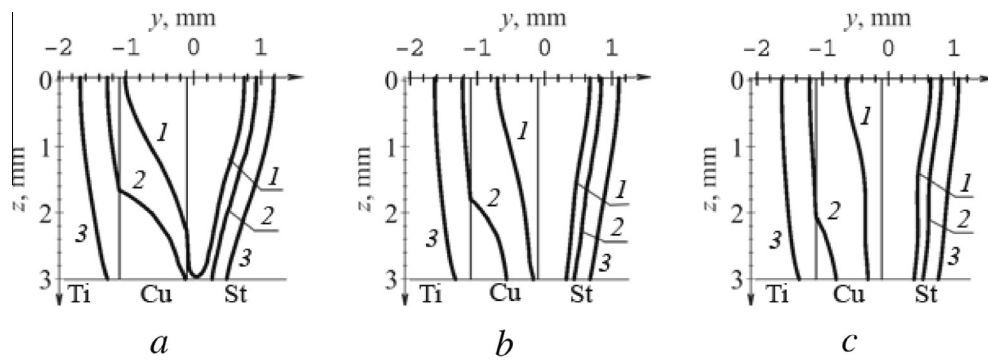


Fig. 6. Position of isotherms in a cross-section of the computation domain at various positions of the beam focus, when the laser operates on the steel plate.

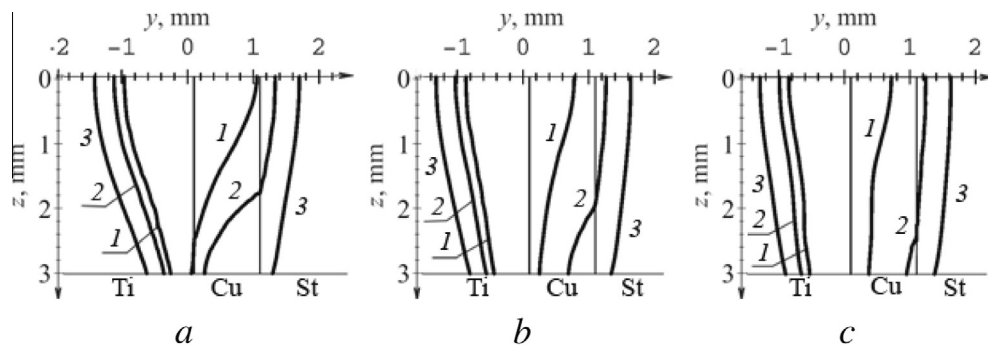


Fig. 7. Position of isotherms in a cross-section of the computation domain at various positions of the beam focus, when the laser operates on the titanium plate.

Fig. 6 shows the results of the isotherms calculation in a cross section of the weld, when the laser beam acts on the steel plate near the joint with the copper insert at various values of deepening of the beam focus: $Z_F = 0$ mm (a), $Z_F = t/2$ (b), and $Z_F = t$ (c). The titanium plate is in region $\{y < -1.1\}$ mm, the copper insert is in strip $\{-1.1\text{ mm} \leq y \leq -0.1\}$ mm, the steel is in region $\{y > -0.1\}$ mm. In this case, just as in the previous one, the copper insert thoroughly melted in all variants of the focal spot location, the steel plate melted to the width of 0.4–0.8 mm. The titanium plate in the case of $Z_F = 0$ mm warmed up to a temperature close to the melting point only in the upper part. Therefore, titanium and steel mixing does not occur here.

Isotherms in the weld joint in the case, when the focal spot of the beam is located at the junction of the titanium plate with the copper insert on the titanium side, are shown in Fig. 7. Titanium plate is located here in region $\{y < 0.1\}$ mm, the copper insert is in strip $\{0.1\text{ mm} \leq y \leq 1.1\}$ mm, and the steel plate is in region $\{y > 1.1\}$ mm. In this case, the titanium plate melts to the width of 0.4–1.0 mm in all the three variants of the focal spot deepening ($Z_F = 0$ mm (a), $Z_F = t/2$ (b), and $Z_F = t$ (c)), the copper insert melts

completely, and the steel plate melts partially in a narrow zone of ~ 1 mm. Hence, here we have the mixing of all three metals and formation of the brittle intermetallic compounds. Therefore, it is possible to predict that the most favorable mode of welding is achieved when the laser beam is directed onto the junction of the steel plate with the copper insert on the steel side. This is confirmed by experimental data, where the greatest strength of the joint occurred in this welding mode [6].

5. Conclusions

A 3D model of thermophysical processes in inhomogeneous materials at laser impact, characterizing features of the temperature fields' distribution in the region of butt welding of titanium and stainless steel using an intermediate copper insert is developed. It is shown that the nature of melting and mixing of the welded materials depends on the position of the focal spot relative to boundaries of the contact and on the spot deepening. The optimal mode corresponds to position of the spot on the steel by its

contact with the copper, while the beam focus is at the bottom surface of the plates welded. Under these conditions, the interaction of elements of the welded materials is significantly reduced, that hinders the processes of formation of brittle intermetallic phases like Ti_xFe_y in the weld joint.

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